

MA 222: Applied Linear programming

Course Code: MA 222

Course Name: Applied Linear programming

L-T-P-C: 3-1-0-4

Prerequisites: Linear algebra

Students intended for: B.Tech. Mathematics & Computing 2nd Year

Elective or Core: Core

Mutual Exclusion: MA 515

Approval: 59th BoA

Course Contents

- **Linear Programming Fundamentals:** Optimization Basics and Origin of LP Problems, Systems of Linear Inequalities, Convex Sets and Feasible Regions. [4 Lectures]
- **Computational Aspects of Linear Programming:** Graphical Method, Simplex Method, Big-M, Dual Problem, Dual Simplex Method, Primal-Dual method. [14 Lectures]
- **Alternative approaches of Linear Programming:** Karmarkar's Algorithm, Transportation problems, Assignment problems. [10 Lectures]
- **Integer Programming:** Introduction to Integer Programming, Gom01y's cutting plane algorithm, Branch and Bound Method, Solving Zero-one problem. [6 Lectures]
- **Game Theory:** Basic Concept and Terminologies, Minimax (Maximin) criterion, Saddle point, Twoperson Zero-sum Game, and Game with Pure and Mixed Strategies, Dominance Principle, Arithmetic Method, and Graphical Method for Solving (2x n) Game, Graphical Method for Solving (m x2) Game and Linear Programming Methods for Rectangular Games. [4 Lectures]

Text books:

1. Luenberger and Ye., **Linear and Nonlinear Programming**, Springer, 2008.
2. Elliott Mendelson and Daniel Zwillinger, **Introducing Game Theory and its Applications**, 2nd Edition, Taylor & Francis, 2004.

References:

1. M.S. Bazaraa, J.J. Jarvis and H.D. Sherali, **Linear Programming and Network Flows**, 2nd Edition, Wiley & Sons, 1990.
2. I. Griva, S.G. Nash, A. Sofer, **Linear and Nonlinear Optimization**, SIAM, 2009.
3. Nocedal and Wright, **Numerical Optimization**, Springer, 2006.
4. Murty, Katta G., ed., **Case Studies in Operations Research: Applications of Optimal Decision Making**, Vol. 212. Springer 2014.
5. Articles as chosen by the instructor.